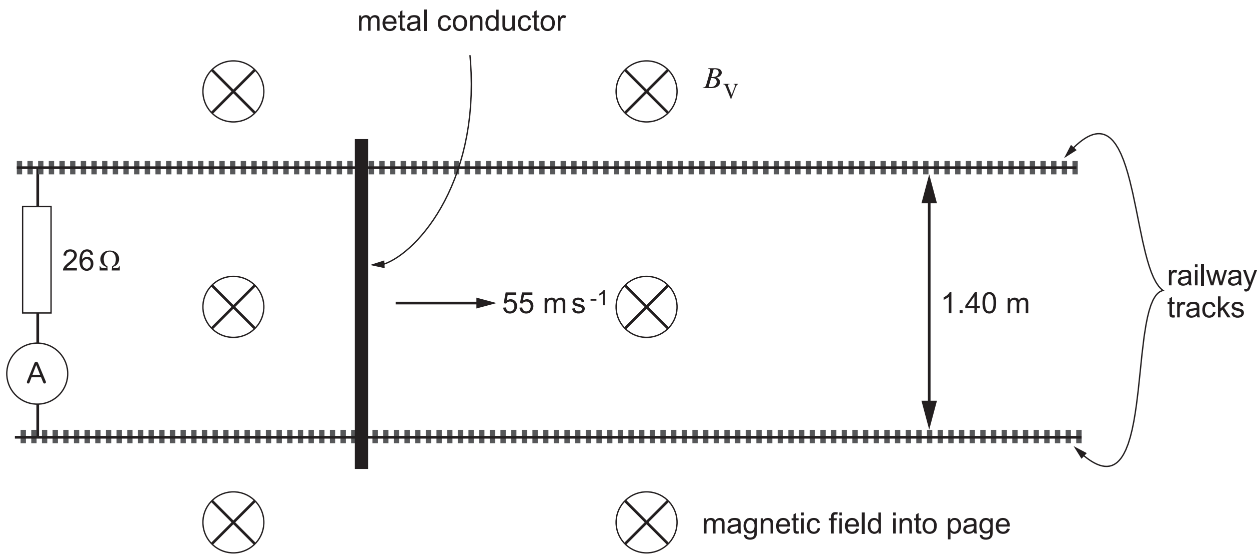
Examiner only

**5.** A metal conductor is placed across horizontal railway tracks and moved quickly in the direction shown. This set up is designed to measure the vertical component ($B_V$) of the Earth's magnetic field.




*(a)* (i) Explain why a current is detected by the ammeter. [2]

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(ii) State why the current is independent of the horizontal component of the Earth's magnetic field. [1]

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*(b)* Calculate a value for the vertical component of the Earth's magnetic field given that the reading on the ammeter is 184 μA. [3]

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Examiner only

(c) State the direction of the current in the resistor and how you obtained this direction. [2]

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(d) A student suggests that the opposing force due to the magnetic field on the moving conductor is negligible compared with other resistive forces. Is the student correct? Justify your answer with a calculation. [4]

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